Monish Prasanna S

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Summary

CS (Cyber Security) student with solid DSA, OOP, and Python fundamentals, plus hands-on experience fine-tuning and red-teaming LLMs and deploying cloud-native apps. Used to testing, breaking, and verifying systems rather than trusting plausible output.

Education

Amrita Vishwa Vidyapeetham — Chennai, India
B.Tech, Computer Science and Engineering (Cyber Security) – CGPA: 7.31 — *Aug 2023 – Expected 2027*

Technical Skills

Languages & Fundamentals: Python, C++, C, JavaScript; DSA, OOP, Operating Systems, Computer Networks

AI / LLM: LoRA/QLoRA fine-tuning (Unsloth, PEFT, TRL), LLM red-teaming, prompt injection, LM Studio, OpenAI-compatible REST APIs

Tools & Cloud: Git, GitHub Actions, Docker, Kubernetes, Linux, Shell Scripting, Cloudflare Workers, Tailscale, REST APIs

Security: Burp Suite, Wireshark, SonarQube (SAST), Malware Analysis, PortSwigger Academy

Projects

Dual-LLM Workbench: Self-Healing Medical LLM | *Python, Llama-3.2-3B, LoRA, Unsloth, LM Studio, JavaScript*

- Built a closed-loop testbed where an uncensored attacker LLM red-teams a LoRA fine-tuned Llama-3.2-3B hospital assistant via prompt injection, role-play jailbreaks, and bulk-extraction queries.
- Wrote a Python leak detector checking responses against 177 PII patterns from a synthetic patient database.
- Automated self-healing: each leak is paired with a refusal (ChatML) and triggers QLoRA retraining (Unsloth, PEFT, TRL; lr 1e-4, 2 epochs); added a web arena (JS + Python proxy) for multi-turn attacker-vs-defender testing.

URL Shortener | *Cloudflare Workers, Cloudflare KV, Edge Routing, Tailwind CSS*

- Deployed a live serverless redirect engine with hash-based key generation, O(1) average-case KV lookups, custom aliases, and click-tracking analytics at zero infrastructure cost.

USB HID Trust Gaps in Modern Operating Systems | *Embedded C/C++, ATtiny85, V-USB*

- Built ATtiny85 HID implants to test physical I/O trust gaps on hardened Windows baselines; automated Wi-Fi credential exfiltration via webhook payload. Presented at ICACSDF 2026.

Web Application Vulnerability Analysis | *Bugcrowd, SonarQube, GitHub Actions, PortSwigger*

- Reported CORS misconfigurations on Bugcrowd; integrated SonarQube SAST into GitHub Actions CI/CD for early detection.

Publication

Evaluating USB HID Trust Gaps in Modern Operating Systems, ICACSDF 2026, UPES Dehradun (IEEE UP Section co-organized), Sept 2026

Certifications

AWS: Certified Cloud Practitioner | **Palo Alto Networks:** Cybersecurity, Network, Cloud & SecOps Fundamentals | **Cisco:** Intro to Cyber Security, Packet Tracer

Volunteering & Leadership

Organizer, Morphx – Amrita CyberNation 2025: designed reverse-engineering and vuln-hunting challenges. **CTF Contributor**, NULLBORN: contributed challenges to the main CTF event.